

Intelligent Traffic Light Control System for Urban Intersections

Development of an artificial intelligence–based traffic signal control system to optimize vehicular flow at urban intersections.

Artificial Intelligence Engineering - Second Semester

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Introduction

Title: Intelligent Traffic Light Control System for Urban Intersections.

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General Objective

Design and implement an algorithmic traffic light control system focused on the intersection of Calle 60 and Carrera 28 (UIS sector), in order to optimize vehicular and pedestrian flow based on real traffic conditions.

Context

In the specific context of the intersection located at Calle 60 and Carrera 28 (UIS sector), traffic congestion and delays are frequently observed, particularly during peak hours. This localized problem highlights how a single intersection can significantly impact overall mobility, increasing waiting times for both vehicles and pedestrians. Traditional traffic light systems at this type of intersection usually operate with fixed timing schemes that do not adapt to real variations in traffic demand. Consequently, green times are often distributed inefficiently, generating unnecessary delays in some directions while others remain underutilized. In addition, pedestrian flow and emergency vehicle priority are not adequately integrated into these conventional systems. In this project, this intersection is not only used as a case study but also as the primary reference environment for the development of the artificial intelligence system. That is, the design, training, and validation of the AI will be directly inspired by the real conditions, behaviors, and traffic patterns observed at this location. By grounding the development in a specific and well-defined scenario, the system can be tailored more precisely to realistic constraints and operational needs. The problem is therefore approached as a dynamic system in which variables such as vehicular flow, pedestrian demand, arrival rates, and waiting times continuously influence decision-making. Based on this, an AI system will be designed to interpret these conditions and autonomously adjust traffic light phases in real time. Through characterization, mathematical modeling, simulation, and evaluation, the project aims to develop a functional and technically viable solution at a small scale, using this intersection as the foundation for the design and testing of the proposed intelligent control system.

1 Research Problem

1.1 Problem Formulation

The problem consists of designing an AI-based traffic light control system capable of optimizing signal timing at the intersection of Calle 60 and Carrera 28. The objective is to minimize vehicle and pedestrian waiting times, reduce congestion levels, and improve overall traffic flow. The system is modeled as a dynamic decision-making problem in which multiple variables interact, including vehicular flow, pedestrian demand, queue lengths, and arrival rates. These variables change over time and directly influence the state of the system. The AI must determine, in real time, the optimal traffic light phases and their duration, subject to operational constraints such as minimum and maximum green times, safety requirements for pedestrians, and priority conditions for emergency vehicles. Formally, the problem can be expressed as the optimization of a performance function that minimizes total delay and queue length, while satisfying system constraints and ensuring efficient and safe traffic operation at the intersection.

1.2 Project Justification

The development of an artificial intelligence (AI)-based traffic light control system for the intersection of Calle 60 and Carrera 28 (UIS sector) is justified by the need to address localized traffic inefficiencies that directly affect mobility at this specific point. Although the problem may appear limited to a single intersection, its impact on vehicle flow, pedestrian safety, and overall traffic performance is significant.

Current traffic light systems operate with fixed timing schemes that do not adapt to real-time traffic conditions. This leads to inefficient use of available road capacity, increased waiting times, and higher levels of congestion, especially during peak hours. Additionally, these systems do not adequately consider variations in pedestrian demand or the need to prioritize emergency vehicles.

By focusing on a specific intersection, this project allows for a more detailed and realistic analysis, enabling the design of a solution tailored to actual traffic patterns and constraints. The implementation of an AI system directly within the traffic light control introduces adaptability, allowing signal timings to respond dynamically to changing conditions.

From a technical perspective, this project integrates concepts from mathematical modeling, optimization, and artificial intelligence, providing a practical application of these disciplines in a real-world scenario. From an operational standpoint, the proposed system aims to reduce delays, improve traffic flow, and enhance the efficiency of the intersection.

1.2 Project Justification

Furthermore, developing and validating the solution at a small scale establishes a foundation for future expansion to more complex traffic networks. Therefore, this project is relevant not only for its immediate local impact but also for its potential scalability and contribution to intelligent urban traffic management systems.

2 Objectives

2.1 Main Objective

Design and implement an intelligent traffic light control system capable of dynamically adjusting signal timing in order to improve vehicular and pedestrian flow at the intersection of Calle 60 and Carrera 28.

2.2 Specific Objectives

Recommended objectives (realistic)

1. Develop a mathematical model that represents the traffic dynamics at the selected intersection.
2. Implement a simulation environment in Python to analyze traffic flow and evaluate different traffic light control strategies.
3. Design a heuristic or AI-based algorithm capable of dynamically adjusting traffic light phases based on traffic conditions.

Additional objectives for evaluation

1. Compare the performance of the proposed system with traditional fixed-time traffic light control.
2. Evaluate system performance using metrics such as queue length, waiting time, and traffic throughput.
3. Analyze the potential scalability of the model for application in larger urban traffic networks.

3 State of the Art

Traffic signal control has become a central topic in intelligent transportation systems due to its direct impact on congestion, travel time, and environmental sustainability. Traditional traffic light systems based on fixed-time control have been widely used; however, they are unable to adapt to dynamic traffic conditions, resulting in inefficient traffic management. To address these limitations, adaptive traffic control systems have been developed, incorporating real-time data from sensors to dynamically adjust signal timings. These systems represent an improvement over static approaches but still rely on predefined rules that may not generalize well under highly variable traffic conditions. In recent years, reinforcement learning (RL) has emerged as one of the most promising approaches for traffic signal control. RL models treat the problem as a sequential decision-making process in which an agent learns optimal control policies through interaction with the environment. This allows traffic lights to adapt autonomously to changing traffic conditions without requiring explicit programming. Several studies have demonstrated the effectiveness of RL-based methods. For example, approaches based on Deep Q-Networks (DQN) dynamically adjust traffic phases using real-time data and have achieved significant reductions in waiting time, queue length, and overall travel time compared to traditional fixed-time systems. More advanced methods include multi-agent reinforcement learning (MARL), where each intersection is modeled as an intelligent agent that cooperates with others. These approaches improve coordination in larger traffic networks and enhance overall system efficiency. Additionally, recent research has explored hierarchical and federated reinforcement learning models, which aim to improve scalability and generalization across different traffic scenarios. These models address challenges such as communication constraints and coordination between multiple intersections. Another important research direction involves the integration of computer vision and real-time data processing, enabling traffic systems to detect vehicles, estimate congestion levels, and improve decision-making accuracy. These technologies enhance the responsiveness of intelligent traffic light systems and support more precise control strategies. Despite these advances, most existing research focuses on large-scale traffic networks. However, recent studies highlight the importance of analyzing individual intersections as dynamic systems, allowing for more precise modeling and efficient localized solutions. In this context, the present project aligns with current research trends by proposing an AI-based traffic light control system applied to a specific intersection. This approach combines dynamic system modeling with intelligent decision-making, contributing to the development of scalable and efficient traffic management solutions.

<https://www.nature.com/articles/s41598-025-91966-1>

<https://www.mdpi.com/2412-3811/10/5/114>

3 State of the Art

<https://link.springer.com/article/10.1186/s40537-025-01104-x>

4 System Visualization

4.1 Environmental Context

The growth of urban traffic in cities like Barrancabermeja has generated significant environmental impacts, mainly due to inefficient traffic flow management. Congestion at intersections, especially those regulated by fixed-time traffic lights, increases the time vehicles remain idle or moving slowly.

One of the main effects is the increase in pollutant emissions, particularly carbon dioxide (CO₂), resulting from fossil fuel combustion. When vehicles idle or frequently stop, fuel consumption rises, increasing emissions.

Additionally, vehicle accumulation in specific areas deteriorates air quality, directly affecting public health. This issue is more severe in densely populated urban zones.

In this context, intelligent traffic control systems provide a solution by optimizing traffic light timing based on real demand, reducing congestion, waiting times, and emissions. This contributes to more sustainable urban systems.

4.2 Economic Context:

Inefficient traffic management also has significant economic impacts. Traditional traffic light systems fail to adapt to real traffic conditions, leading to poor resource utilization. Key costs include increased fuel consumption due to constant stopping and starting, as well as higher vehicle maintenance costs caused by inefficient driving conditions.

Traffic congestion also results in productivity losses due to wasted time for drivers and transportation services, increasing operational costs.

Another important consequence is reduced efficiency in critical services such as emergency transport, where delays can have serious implications.

Optimizing traffic light control through intelligent systems improves resource utilization, reduces fuel consumption, lowers costs, and enhances overall transportation efficiency.

4.3 Academic Justification

This project integrates fundamental concepts from computer science and mathematics to solve a real-world traffic problem.

Graph theory is used to model intersections as nodes and roads as edges, allowing analysis of traffic flow using variables such as vehicle count and waiting time.

Heuristic algorithms are applied for real-time decision-making, providing efficient solutions to a computationally complex problem.

4 System Visualization

Additionally, computational simulation is used to evaluate system performance under different traffic scenarios, analyzing metrics such as waiting time and congestion. Finally, the model is implemented in Python, enabling a flexible and scalable simulation environment.

5 Mathematical and Computational Modeling

5.1 Definición formal del sistema dinámico

Desde un enfoque matemático, el comportamiento del sistema puede describirse mediante la siguiente ecuación de evolución:

$$X(t+1) = f(X(t), U(t), \xi(t)) \quad (5.1)$$

donde $X(t)$ representa el vector de estado del sistema en el instante t , $U(t)$ corresponde al conjunto de decisiones de control asociadas al semáforo, y $\xi(t)$ agrupa las perturbaciones externas, principalmente las llegadas de vehículos.

Esta formulación permite entender que el estado futuro del sistema depende tanto de su condición actual como de las decisiones tomadas y de factores externos no controlables.

5.2 Variables del modelo

El modelo se construye a partir de la definición de variables que permiten describir, controlar y analizar el sistema.

5.2.1 Variables de estado

Para cada vía i se definen las siguientes variables:

- $Q_i(t)$: número de vehículos en cola, el cual representa el nivel de congestión en la vía.
- $W_i(t)$: tiempo de espera acumulado de los vehículos, utilizado como indicador del costo temporal asociado a la congestión.
- $T_i(t)$: tiempo durante el cual la vía ha permanecido sin recibir servicio (luz verde).

La evolución de la longitud de la cola se modela mediante:

$$Q_i(t+1) = Q_i(t) + \lambda_i(t) - \mu_i(t) \quad (5.2)$$

donde $\lambda_i(t)$ es la tasa de llegada de vehículos y $\mu_i(t)$ la tasa de salida.



Figure 5.1: Geographical Location of the Study Area: Intersection of 60th Street and 28th Avenue

5.2.2 Variables de control

Las variables de control representan las decisiones tomadas por el sistema:

- $S_i(t) \in \{0, 1\}$: indica si la vía i tiene luz verde en el instante t .
- $G(t)$: duración de la fase verde asignada.

Se establece la siguiente restricción fundamental:

$$\sum_i S_i(t) = 1 \quad (5.3)$$

lo que implica que únicamente una vía puede recibir luz verde en un instante determinado.

5.2.3 Variables externas

El modelo también considera variables externas que influyen en el comportamiento del sistema:

- $A_i(t) \in \{0, 1\}$: indica la presencia de un vehículo de emergencia en la vía i .
- $P_i(t)$: número de peatones en espera.

Estas variables no son controladas directamente, pero afectan las decisiones del sistema.

5.3 Modelado del flujo vehicular

El comportamiento del flujo vehicular se describe a partir de dos componentes fundamentales: la llegada y la salida de vehículos.

La tasa de llegada $\lambda_i(t)$ representa el número de vehículos que ingresan a la vía durante el intervalo de tiempo Δt .

Por otro lado, la tasa de salida depende del estado del semáforo:

$$\mu_i(t) = S_i(t) \mu_{\max} \quad (5.4)$$

donde $\mu_{\max}$ representa la capacidad máxima de evacuación de la vía.

El tiempo de espera acumulado evoluciona según:

$$W_i(t+1) = W_i(t) + Q_i(t) \quad (5.5)$$

lo que implica que cada vehículo en cola contribuye al incremento del tiempo total de espera.

5.4 Restricciones del sistema

El modelo incorpora diversas restricciones que garantizan su coherencia operativa y seguridad.

- Solo una fase puede estar activa en cada instante.
- Existe un tiempo mínimo de luz verde $G_{\min}$.
- Se considera una fase de transición (luz amarilla).
- Se evita la conmutación inmediata entre fases incompatibles.

Para vehículos de emergencia:

$$A_k(t) = 1 \Rightarrow S_k(t) = 1 \quad (5.6)$$

5.5 Función objetivo del sistema

Se define una función de costo global:

$$J = \sum_i (\alpha Q_i(t) + \beta W_i(t) + \gamma T_i(t)) \quad (5.7)$$

donde α , β y γ son ponderaciones asociadas a congestión, espera y tiempo sin servicio.

5.6 Estrategia de control heurística

Se define una función de prioridad para cada vía:

$$\pi_i(t) = \alpha Q_i(t) + \beta W_i(t) + \gamma T_i(t) \quad (5.8)$$

La decisión de control es:

$$\text{Seleccionar } i^* = \arg \max_i \pi_i(t) \quad (5.9)$$

El tiempo de luz verde se ajusta dinámicamente:

$$G(t) = G_{\min} + k Q_i(t) \quad (5.10)$$

5.7 Algoritmo de operación

En cada instante t :

1. Actualizar $\lambda_i(t)$
2. Calcular $Q_i(t)$
3. Actualizar $W_i(t)$ y $T_i(t)$
4. Evaluar $\pi_i(t)$
5. Seleccionar vía activa
6. Aplicar fase verde
7. Actualizar $\mu_i(t)$

5.8 Métricas de desempeño

- Tiempo promedio de espera
- Longitud promedio de cola
- Flujo vehicular total
- Tiempo de respuesta a emergencias

5.9 Análisis del modelo

El modelo presenta comportamiento determinista en la toma de decisiones, con variabilidad en las entradas. Tiene complejidad computacional lineal respecto al número de vías y es escalable a redes más grandes.

5.10 Ventajas del enfoque

El enfoque destaca por:

- Simplicidad
- Bajo costo computacional
- Facilidad de implementación
- Capacidad de simulación sin infraestructura física

6 Graph-Based Modeling of the Traffic System

6.1 Graph Definition

The urban traffic system is modeled as a directed and weighted graph:

$$G = (V, E)$$

where:

- V : set of nodes, representing intersections controlled by traffic lights.
- E : set of edges, representing the segments connecting intersections.

6.2 Graph Characteristics

The graph has the following properties:

- **Directed:** each edge has a defined direction, corresponding to traffic flow.
- **Weighted:** each edge $e \in E$ has an associated weight $w(e)$, which may represent:
 - travel time
 - road length
 - congestion level
 - vehicle flow

6.3 Associated Variables

The following variables are defined:

- $x_i(t)$: number of vehicles at node i at time t
- $f_{ij}(t)$: flow of vehicles from node i to node j
- $u_i(t)$: traffic light state at node i
 - 0 = red
 - 1 = green

6.4 System Dynamics on the Graph

The evolution of traffic at each node is given by:

$$x_i(t+1) = x_i(t) + \sum_{j \in V} f_{ji}(t) - \sum_{j \in V} f_{ij}(t)$$

This equation represents the balance between incoming and outgoing vehicles at each intersection.

6.5 Traffic Light Control Relationship

The flow $f_{ij}(t)$ depends on the traffic light state:

$$f_{ij}(t) = \begin{cases} \mu_{ij} \cdot x_i(t), & \text{if } u_i(t) = 1 \\ 0, & \text{if } u_i(t) = 0 \end{cases}$$

where μ_{ij} is the vehicle departure rate.

6.6 Matrix Representation

The system can also be expressed in matrix form:

$$X(t+1) = X(t) + A \cdot F(t)$$

where:

- $X(t)$: state vector (vehicles at each node)
- $F(t)$: flow vector
- A : graph incidence matrix

6.7 Objective Function

The optimization problem is defined as:

$$\min \sum_{i \in V} x_i(t)$$

The goal is to minimize:

- total congestion
- waiting times
- vehicle accumulation

6.8 Advantages of the Graph Approach

- Realistic representation of the road network
- Enables the use of optimization algorithms
- Facilitates computational simulation
- Scalable to multiple intersections

6.9 Prototype Implementation

For implementation, the graph can be modeled using Python libraries such as `networkx`, where:

- Nodes represent intersections
- Edges represent roads

Simulations can include:

- traffic light switching
- vehicle flow dynamics
- congestion reduction analysis

7 Conclusions (Preliminary)

The graph-based modeling approach provides a structured and mathematically consistent framework for representing and analyzing urban traffic systems. By defining intersections as nodes and road segments as directed, weighted edges, it becomes possible to capture the dynamic behavior of vehicle flow in a clear and scalable manner.

The incorporation of traffic light control variables into the graph model allows for the interaction between infrastructure and traffic dynamics to be explicitly represented. This enables the formulation of optimization problems aimed at reducing congestion, minimizing waiting times, and improving overall traffic efficiency.

Furthermore, the use of a mathematical and computational framework facilitates the implementation of simulation tools, which are essential for testing different control strategies before real-world deployment. This not only improves decision-making but also reduces operational risks.

In conclusion, the integration of graph theory with dynamic system modeling constitutes an effective and flexible approach for the design of intelligent traffic light systems, with strong potential for practical applications in modern urban environments.

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